

Vanshay Gupta

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EDUCATION

University of Southern California, Los Angeles, California

May 2024

Master of Science in Computer Science

Relevant Coursework: Analysis of Algorithms, Machine Learning, Multimedia Systems Design, Operating Systems

Dwarkadas J. Sanghvi College of Engineering, University of Mumbai, India

Jun 2022

Bachelor of Engineering in Computer Engineering

(CGPA: 9.47/10)

TECHNICAL SKILLS

- Programming Languages: Python, C++, SQL, C, R, Java
- Web Technologies: JavaScript, PHP, Node.js, REST, React.js, Flask, TypeScript, Django, Angular, Next.js
- Software: Postgres, MongoDB, AWS, Heroku, Git, Selenium, Docker, Tableau, Hadoop, Keras, OpenCV, PyTorch, Spark

WORK EXPERIENCE

University of Southern California – Melady Lab (Viterbi School of Engineering)

Los Angeles, California

Graduate Student Researcher | Machine Learning, Time Series, Python, Transformers

May 2023–Present

- Developing Python package to automate and standardize time series model execution and analysis for benchmark datasets
- Analyzed performance of linear and transformer based models for short-term and long-term time series forecasting

University of Southern California – Marshall School of Business

Los Angeles, California

Research Intern | Machine Learning, NLP, Web Crawling, Time Series, GPT-4, Sentiment Analysis

May 2023–Sep 2023

- Examined dynamics between restaurant pricing, market conditions and customer perception by analyzing changes in Yelp user reviews with increases in menu prices over a period of 10 years
- Explored companies' future outlook and risk embracement in shareholder communication in correlation with stock market movements and impact of COVID-19 on business policies by forming time series data from financial reports
- Extracted annual financial filings of Fortune 100 companies for 20 years using Selenium and subsequently utilized GPT-4 to generate numerical ratings on a scale of 10 for yearly shareholder letters across eight sentiment dimensions

University of Southern California – CKIDS (Viterbi School of Engineering)

Los Angeles, California

Graduate Student Researcher | Machine Learning, NLP, Python, Gensim, Topic Modeling

Sep 2022–Dec 2022

- Reduced reported police misconduct cases by 35% by identifying common traits in these reports to generate awareness and improve academy training programs, along with implementing an encrypted database in collaboration with Microsoft
- Extracted relevant articles using keyword matching with Goose3, followed by abstractive summarization with PEGASUS
- Evaluated similarity with existing reports in USC Police Misconduct Registry to categorize articles using LDA and LSI

Julius Baer Group

Mumbai, India

Software Engineer Intern | Data Engineering, Web Development, Python, SQL, Node.js, Flask

Jun 2021–Aug 2021

- Improved efficiency of relationship managers by 10% by developing a news alert tool that saved weekly manual effort of 4 hours, using Cheerio, MySQL, and Node.js to aggregate customized and summarized client updates periodically
- Implemented a standard data engineering pipeline to automatically analyze and update bond valuation data procured from multiple credit rating agency websites, reducing average time by 97.5% (30+ minutes to 45 seconds)

ANAROCK Property Consultants

Mumbai, India

Web Development Intern | React.js, Next.js, TypeScript, AWS

Jul 2020–Sep 2020

- Led development of company's technology website in TypeScript and SASS, resulting in a 20% increase in customer conversion through the registration portal, and deployed using Amazon ELBs to distribute traffic across EC2 instances
- Streamlined company website resulting in 30% SEO score enhancement, and built new pages with React.js and Next.js

PROJECTS

VizScan: Instant Query Video Searching

[DOCS]

- Achieved 600x speedup over standard video searching methods by designing a Python algorithm to compute average hash of video queries and search for most similar segment from a database of hash-indexed full-length videos

Insurance Eligibility and Premium Computation

[DOCS]

- Designed web portal to automate medical underwriting by calculating risk factor and insurance premiums based on holistic applicant health profiles and weights assigned to diseases from frequency in hospital records
- Utilized Flask and Firebase to build scalable disease prediction algorithm with 93% accuracy and deployed on Heroku

University Career Portal

[DOCS]

- Spearheaded development of comprehensive career portal to modernize and streamline on-campus job applications using React.js and Django, enhancing efficiency for students and employers through 15% reduction in processing time
- Implemented job tracking module to increase accessibility, augmenting application submissions by 10% across majors

PUBLICATIONS

Time Series Analysis and Forecasting of Air Quality in India

[LINK]

2023 Fifth International Conference on Electrical, Computer and Communication Technologies (ICECCT). IEEE, 2023

Data-driven Preprocessing Techniques for Early Diagnosis of Diabetes, Heart and Liver Diseases

[LINK]

2021 Fourth International Conference on Electrical, Computer and Communication Technologies (ICECCT). IEEE, 2021